

Problem 1 – Question 6

Measurement

6(a): Joint measurement of Z_1Z_2 and X_1X_2

The two observables commute:

$$(Z_1Z_2)(X_1X_2) = (Z_1X_1)(Z_2X_2) = (X_1Z_1)(X_2Z_2) = (X_1X_2)(Z_1Z_2).$$

They can therefore be measured jointly without an uncertainty conflict.

Z_1Z_2 syndrome. Prepare an ancilla a_Z in $|0\rangle$, apply

$$\text{CNOT}_{q_1 \rightarrow a_Z}, \quad \text{CNOT}_{q_2 \rightarrow a_Z},$$

and measure a_Z in the Z basis. If the measured bit is m_Z , the measured eigenvalue is

$$z = (-1)^{m_Z}.$$

The data state is projected by

$$\Pi_z^{(ZZ)} = \frac{\mathbb{I} + zZ_1Z_2}{2}.$$

X_1X_2 syndrome. Prepare a second ancilla a_X in $|+\rangle$ by applying H to $|0\rangle$. Apply

$$\text{CNOT}_{a_X \rightarrow q_1}, \quad \text{CNOT}_{a_X \rightarrow q_2},$$

then apply H to a_X and measure it in the Z basis. If the result is m_X , the measured eigenvalue is

$$x = (-1)^{m_X},$$

and the data state is projected by

$$\Pi_x^{(XX)} = \frac{\mathbb{I} + xX_1X_2}{2}.$$

The two syndrome extractions may be done sequentially with one resettable ancilla, or with two ancillas.

Conditioned on outcomes $m_Z = m_X = 0$, the normalized output is

$$|\psi_{++}\rangle = \frac{(\mathbb{I} + Z_1Z_2)(\mathbb{I} + X_1X_2)|\psi\rangle}{\|(\mathbb{I} + Z_1Z_2)(\mathbb{I} + X_1X_2)|\psi\rangle\|}.$$

This is the joint $+1$ projection. The success probability is

$$p_{++} = \langle\psi| \frac{(\mathbb{I} + Z_1Z_2)(\mathbb{I} + X_1X_2)}{4} |\psi\rangle.$$

For two data qubits, the joint $+1$ eigenspace is one-dimensional and is spanned by

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

If deterministic preparation of $|\Phi^+\rangle$, rather than literal postselection, is desired, use classical feedback: apply X_1 when $m_Z = 1$, and apply Z_1 when $m_X = 1$. X_1 flips only the ZZ eigenvalue, while Z_1 flips only the XX eigenvalue.

6(b): Projective measurement of a general Pauli operator

Let

$$P = \eta P_1 \otimes \cdots \otimes P_n, \quad \eta \in \{+1, -1\}, \quad P_j \in \{\mathbb{I}, X, Y, Z\}.$$

Ignore identity positions. On every nonidentity position, rotate its Pauli operator to Z :

$$B_j = \begin{cases} H, & P_j = X, \\ HS^\dagger, & P_j = Y, \\ \mathbb{I}, & P_j = Z. \end{cases} \quad B_j P_j B_j^\dagger = Z.$$

Chronologically, Y -basis conversion applies $S^\dagger$ and then H .

Prepare an ancilla a in $|0\rangle$. After the basis changes, apply a CNOT from every nonidentity data qubit to a , then measure a in the Z basis. The measured bit m gives eigenvalue

$$p = \eta(-1)^m.$$

Finally undo the basis changes if the data must remain available. The corresponding projectors are

$$\boxed{\Pi_\pm = \frac{\mathbb{I} \pm P}{2}}.$$

This is a nondestructive projective measurement: within a fixed eigenspace, superpositions are preserved. It uses one ancilla and $\text{wt}(P)$ CNOTs. If destroying the data is acceptable, one may instead measure every nonidentity qubit in its local Pauli basis and multiply the individual eigenvalues.

6(c): Reducing the GHZ-like state to one qubit

Write

$$|\Psi_n\rangle = \alpha|0\rangle^{\otimes n} + \sqrt{1 - \alpha^2}|1\rangle^{\otimes n}.$$

Keep q_1 . Measure $q_2, \dots, q_n$ in the X basis by applying Hadamards and then measuring in the computational basis. Let the outcomes be $m_2, \dots, m_n$, and define their parity

$$p = m_2 \oplus \cdots \oplus m_n.$$

Because

$$\langle \pm | 0 \rangle = \frac{1}{\sqrt{2}}, \quad \langle \pm | 1 \rangle = \frac{\pm 1}{\sqrt{2}},$$

the unmeasured qubit is left in

$$\alpha|0\rangle + (-1)^p \sqrt{1 - \alpha^2}|1\rangle.$$

Apply the classically controlled correction Z^p to q_1 . The final state is

$$\boxed{\alpha|0\rangle + \sqrt{1 - \alpha^2}|1\rangle}.$$

Every outcome string is acceptable; the parity correction removes its known phase. Thus the protocol succeeds deterministically and requires no ancillary qubits.